

The Griffiths exponent of random-field Ising dynamics at strong Gaussian disorder

Abstract

We determine the double-logarithmic Griffiths exponent for the random-field Ising model on $\mathbb{Z}^d$, $d \geq 2$, at fixed inverse temperature β . We consider free-boundary boxes of side order n , independent centered Gaussian fields, and continuous-time heat-bath dynamics with rate-one updates per site. Assume that, along increasing side lengths $m_j \rightarrow \infty$ with $m_{j+1}/m_j \rightarrow 1$, the zero-field model admits fixed spin boundary conditions with relaxation time at least $\exp\{cm_j^{d-1}\}$ for some $c > 0$. For every sufficiently large fixed field variance, the relaxation time and the worst-initial-state total-variation mixing time at error $1/4$ both have scale $\exp\{(\log n)^{(d-1)/d+o_{\mathbb{P}_h}(1)}\}$, where $o_{\mathbb{P}_h}(1)$ denotes convergence to zero in probability over the fields. Under the same pure-model hypothesis, the lower bound holds at every fixed positive variance and attains the power predicted by El Alaoui, Eldan, Gheissari, and Piana (Ann. Probab. 2026). We embed a pure-system slow mode in a weak-field island with an evolving high-field shell. A test function independent of the shell reduces the comparison to equilibrium defect moments, allowing fixed-strength shell fields. The upper bound simulates disorder-adapted block updates by parallel local dynamics and applies censoring. In dimension two, the pure-model hypothesis holds for every $\beta > \beta_c(2)$.

Note. This paper was generated entirely by AI, including MiMo, using an automated research pipeline developed by Chenghua Liu and Hanyu Li.

1 Introduction

Strong random fields suppress the influence of neighboring spins at most sites of an Ising system. Nevertheless, a large sample contains regions where the fields are atypically weak and low-temperature collective behavior can survive. A box of side m that retains a pure-system free-energy barrier of surface order can take $\exp\{cm^{d-1}\}$ time to relax. The finite-volume time scale therefore depends on how large such a box is likely to be. This competition between a rare region's probability cost and the lifetime of its slow mode is the Griffiths mechanism; it allows slow dynamics to coexist with rapid decay of correlations in typical environments.

Cesi, Maes, and Martinelli [2] developed rare-region estimates for disordered spin dynamics with random interactions. For the random-field Ising model, El Alaoui, Eldan, Gheissari, and Piana [3] proved that the relaxation time is subpolynomial in the volume under strong spatial mixing in expectation, or under an explicit strong-disorder condition. Their Theorem 5.2 gives, on $\Lambda_n = [-n, n]^d \cap \mathbb{Z}^d$,

$$\text{gap}_{\Lambda_n}^{-1} \leq \exp\left\{C(\log n)^{(d-1)/d}(\log \log n)^{d-1}\right\} \quad (1.1)$$

with high probability. At sufficiently low temperature, their Proposition 6.1 gives a mixing-time lower bound with power $(d-1)/(4d)$ of $\log n$, and they predict the power $(d-1)/d$. In two dimensions, this leaves powers $1/8$ and $1/2$ in the exponential. The unresolved point is whether rare regions quantitatively account for the scale of the upper bound.

Embedding a slow pure box is delicate because its boundary spins are part of the surrounding dynamics. One natural approach is to favor a prescribed boundary by large fields and require the shell to remain in that configuration until the interior relaxes. A shell spin has conditional defect

probability at most $q_H \asymp e^{-2H}$ at fixed temperature. Thus a union bound over updates up to time T incurs the factor $|S|Tq_H$, where S is the shell. At the target time $T = \exp\{cm^{d-1}\}$, this estimate forces the field threshold H to grow with the surface scale. For Gaussian fields, such a requirement makes the shell increasingly expensive as a disorder event. Weak fields in the interior alone do not resolve this boundary problem.

We use a stationary comparison that permits shell defects. Extend a pure-box gap eigenfunction to the full system by making it independent of every spin outside the box. Updates of the shell then contribute no Dirichlet energy. Conditioning on the shell compares both the energy and variance of this function with their pure-system values; the resulting error involves exponential moments of the equilibrium number of defects. A sufficiently large *fixed* shell-field threshold controls these moments on the surface scale. Inside the box, fields of size $O(m^{-1})$ produce a total perturbation of the same order. The island event has logarithmic probability cost $O(m^d \log m)$, while its relaxation barrier remains of order m^{d-1} . Balancing the cost against the number of possible locations gives

$$m \asymp (\log n / \log \log n)^{1/d}, \quad \log t_{\text{rel}}(n, h) \gtrsim (\log n / \log \log n)^{(d-1)/d}.$$

This proves the predicted lower-bound exponent whenever the pure model supplies a fixed-boundary slow box.

To obtain the same exponent for mixing from every initial configuration, we also need a quantitative upper bound in total variation. Applying the usual spectral bound to the entire volume would multiply (1.1) by a factor of at least volume order through the minimum stationary mass. Instead, we use the disorder-dependent blocks of the strong-disorder argument in [3] and mix only within those blocks. Overlap makes an initial discrepancy more likely to be removed than to propagate across a block boundary. A coloring permits disjoint block chains to run simultaneously, and the censoring theorem of Peres and Winkler [5] compares the resulting schedule with the full dynamics. This confines the spectral-to-mixing loss to polylogarithmic-size blocks. The relaxation upper bound is due to [3]; the new lower bound and the direct worst-start comparison together identify the exponent. Han [4] obtains polynomial mixing under anti-concentration on general bounded-degree graphs; the result here identifies the exponent associated with the lattice surface scale.

1.1 Model and main result

Fix $d \geq 2$, $\beta > 0$, and $\sigma \in (0, \infty)$. Realize $h = (h_x)_{x \in \mathbb{Z}^d}$ as independent $N(0, \sigma^2)$ variables. On $\Lambda_n = [-n, n]^d \cap \mathbb{Z}^d$, with free boundary conditions, let

$$\mu_{\Lambda_n, h}(\eta) = \frac{1}{Z_{\Lambda_n, h}} \exp \left\{ \beta \sum_{\substack{\{x, y\} \subset \Lambda_n \\ x \sim y}} \eta_x \eta_y + \sum_{x \in \Lambda_n} h_x \eta_x \right\}. \quad (1.2)$$

Here $x \sim y$ means nearest-neighbor adjacency. Throughout, Glauber dynamics means continuous-time heat-bath dynamics with an independent rate-one clock at each vertex. Write

$$t_{\text{rel}}(n, h) = \text{gap}_{\Lambda_n}(h)^{-1}, \quad t_{\text{mix}}(\varepsilon; n, h) = \inf \left\{ t : \max_{\eta} \|P_t^h(\eta, \cdot) - \mu_{\Lambda_n, h}\|_{\text{TV}} \leq \varepsilon \right\}.$$

Probabilities concerning the environment are denoted by $\mathbb{P}_h$. All logarithms are natural. Constants $c, C > 0$ may change from line to line and depend only on fixed parameters, unless specified otherwise.

For $D \subset V \subseteq \mathbb{Z}^d$, set

$$\partial_V D = \{y \in V \setminus D : \text{there exists } x \in D \text{ with } x \sim y\}, \quad \partial_{\text{ext}} D = \partial_{\mathbb{Z}^d} D.$$

All boundaries below are external vertex boundaries. A fixed boundary condition contributes $\beta \eta_x \phi_y$ for each edge from an interior vertex x to a fixed boundary vertex y . We write $\mu_{D,h}^\phi$ for the resulting measure. When $D \subset \Lambda_n$ is updated conditionally inside (1.2), its boundary is $\partial_{\Lambda_n} D$; there are no interactions with vertices outside Λ_n .

The lower bound uses a pure-system slow mode that can be imposed by a fixed spin boundary. Let $B_m = [1, m]^d \cap \mathbb{Z}^d$, and let $\nu_m^{\zeta_m}$ be the zero-field Ising measure in B_m with fixed boundary condition $\zeta_m \in \{-1, +1\}^{\partial_{\text{ext}} B_m}$.

Assumption 1.1 (Surface-slow pure boxes). There are $c_0 > 0$ and an increasing sequence $\mathcal{M} = \{m_1 < m_2 < \dots\}$ with $m_{k+1}/m_k \rightarrow 1$, such that, for every sufficiently large $m \in \mathcal{M}$, a boundary condition ζ_m can be chosen with

$$\text{gap}(\nu_m^{\zeta_m}) \leq e^{-c_0 m^{d-1}}. \quad (1.3)$$

The upper bound uses the following explicit condition, which is the Gaussian specialization of Assumption 5.1 in [3].

Assumption 1.2 (Strong disorder). There exists $K > 0$ such that

$$\rho := \frac{1}{1 + e^{2(K-2d\beta)}} < \frac{1}{40d}, \quad p_K := \mathbb{P}_h(|h_0| \leq K) < \frac{1}{40d}. \quad (1.4)$$

The first inequality controls the error at a strong-field site; the second makes weak-field sites sparse. Both hold at sufficiently large Gaussian variance for any fixed d, β .

Theorem 1.3 (Relaxation and worst-start mixing). *Assume Assumptions 1.1 and 1.2, and put $a = (d-1)/d$. There exist $c, C > 0$ such that, with $\mathbb{P}_h$ -probability tending to one as $n \rightarrow \infty$,*

$$\exp\left\{c \left(\frac{\log n}{\log \log n}\right)^a\right\} \leq Y_n \leq \exp\left\{C(\log n)^a (\log \log n)^{d-1}\right\} \quad (1.5)$$

simultaneously for $Y_n = t_{\text{rel}}(n, h)$ and $Y_n = t_{\text{mix}}(1/4; n, h)$. Consequently, for either choice,

$$\frac{\log \log \max\{e, Y_n\}}{\log \log n} \xrightarrow{\mathbb{P}_h} \frac{d-1}{d}. \quad (1.6)$$

The constants may depend on the fixed parameters and on the pure-system input, but not on n . The truncation in (1.6) only defines the double logarithm away from the high-probability event in (1.5).

The lower estimate uses only Assumption 1.1 and holds for every fixed $\sigma > 0$. The upper estimate uses only Assumption 1.2. Thus the theorem separates the pure-system input that creates a slow island from the disorder condition that controls relaxation elsewhere. In $d \geq 3$, the pure-box property remains an explicit hypothesis; the unconditional low-temperature application below is two-dimensional. Our worst-start argument uses the stated strong-disorder condition, rather than the larger parameter range described by strong spatial mixing in expectation.

The exponent in (1.6) is a double-logarithmic one. The factors of $\log \log n$ in (1.5) are not matched: the lower bound pays for fields of size $O(m^{-1})$ throughout an island, while the upper bound pays for a covering by blocks adapted to the disorder. The theorem does not determine these corrections or a leading constant.

An explicit, nonoptimal Gaussian threshold is useful for stating a concrete consequence. Set

$$K_* = 2d\beta + \frac{1}{2} \log(80d), \quad \sigma_0(d, \beta) = 80dK_*. \quad (1.7)$$

For $\sigma \geq \sigma_0(d, \beta)$, choosing $K = K_*$ gives

$$\rho = \frac{1}{1 + 80d} < \frac{1}{40d}, \quad p_K \leq \sqrt{\frac{2}{\pi}} \frac{K_*}{\sigma} < \frac{1}{40d}.$$

Thus Assumption 1.2 holds for every sufficiently large fixed σ .

Corollary 1.4 (Two-dimensional low-temperature model). *For $d = 2$, every $\beta > \beta_c(2)$, and every fixed $\sigma \geq \sigma_0(2, \beta)$, the bounds and limits of Theorem 1.3 hold with exponent $1/2$.*

Proof. Alexander's Theorem 1.1(ii) [1] supplies fixed $\{-1, +1\}$ -valued boundary conditions on $[-N, N]^2 \cap \mathbb{Z}^2$ with spectral gap at most $Ce^{-\lambda \min\{k, N-k\}}$, whenever $\min\{k, N-k\} \geq K_\beta \log N$. Take $k = \lfloor N/2 \rfloor$ and translate the square to B_{2N+1} . Absorbing the prefactor into the exponential gives Assumption 1.1 along the odd integers. Their consecutive ratios tend to one. Equation (1.7) gives Assumption 1.2. $\square$

Section 2 proves the lower bound through the shell comparison and the island probability estimate. Section 3 constructs the blocks and their parallel simulation. The elementary spectral tools used in both arguments are collected in Appendix A.

2 Rare regions and the relaxation lower bound

The lower bound is obtained from a single test function supported on one island. We first compare its Rayleigh quotient with that of a pure box, uniformly over the fields outside the island and its shell. Once this deterministic comparison is available, independent placements of the island supply the probability estimate.

Proposition 2.1 (Lower bound at every fixed Gaussian variance). *Under Assumption 1.1, for every fixed $\sigma \in (0, \infty)$, there is $c > 0$ such that, with $\mathbb{P}_h$ -probability tending to one,*

$$\min\{t_{\text{rel}}(n, h), t_{\text{mix}}(1/4; n, h)\} \geq \exp\left\{c \left(\frac{\log n}{\log \log n}\right)^{(d-1)/d}\right\}. \quad (2.1)$$

For a Gibbs measure π on a finite spin set V , the rate-one heat-bath Dirichlet form is

$$\mathcal{E}_\pi(f, f) = \sum_{x \in V} \mathbb{E}_\pi \left[\text{Var}_\pi(f \mid \eta_{V \setminus \{x\}}) \right], \quad \text{gap}(\pi) = \inf_{\text{Var}_\pi(f) > 0} \frac{\mathcal{E}_\pi(f, f)}{\text{Var}_\pi(f)}.$$

The comparison below transfers a slow mode from a box with fixed boundary conditions to a larger system in which the same boundary spins evolve.

Let B be a finite box, $S = \partial_{\text{ext}} B$, and $\zeta \in \{-1, +1\}^S$. Let ν be the zero-field Ising measure in B with boundary ζ . Let π be an Ising measure on a finite domain $\Omega \supset B \cup S$, with arbitrary fields and arbitrary fixed boundary conditions outside Ω . Define

$$A_B = \sum_{x \in B} |h_x|, \quad k(\tau) = |\{y \in S : \tau_y \neq \zeta_y\}|, \quad b = 4\beta \max_{y \in S} |\{x \in B : x \sim y\}| \leq 8d\beta.$$

Lemma 2.2 (Soft-shell comparison). *With the notation above,*

$$\text{gap}(\pi) \leq \text{gap}(\nu) e^{4A_B} \frac{\mathbb{E}_\pi e^{bk(\eta_S)}}{\mathbb{E}_\pi e^{-bk(\eta_S)}}. \quad (2.2)$$

Proof. Let f be a gap eigenfunction of ν , extended to Ω by ignoring spins outside B . This choice eliminates all exterior updates from $\mathcal{E}_\pi(f, f)$, including updates of the shell. We compare its conditional energy and variance for each shell configuration, and then average those comparisons under π .

Condition on $\eta_{\Omega \setminus B}$, and let μ_τ be the conditional law in B , where $\tau = \eta_S$. Its logarithmic tilt relative to ν , before normalization, is

$$U_\tau(\eta) = \sum_{x \in B} h_x \eta_x + \beta \sum_{\substack{x \in B, y \in S \\ x \sim y}} \eta_x (\tau_y - \zeta_y).$$

The field contribution has oscillation at most $2A_B$, and each edge incident to a shell defect contributes at most 4β . Consequently,

$$\text{osc } U_\tau \leq 2A_B + bk(\tau) =: D_\tau, \quad e^{-D_\tau} \leq \frac{d\mu_\tau}{d\nu} \leq e^{D_\tau}.$$

The variational characterization of variance gives

$$\text{Var}_{\mu_\tau}(f) = \inf_c \mu_\tau[(f - c)^2] \geq e^{-D_\tau} \text{Var}_\nu(f). \quad (2.3)$$

For configurations η, η' differing at one site, the heat-bath conductance of a measure μ is

$$q_\mu(\eta, \eta') = \frac{\mu(\eta)\mu(\eta')}{\mu(\eta) + \mu(\eta')}.$$

If $\mu = r\nu$, then

$$\frac{q_\mu(\eta, \eta')}{q_\nu(\eta, \eta')} = \frac{r(\eta)r(\eta')(\nu(\eta) + \nu(\eta'))}{r(\eta)\nu(\eta) + r(\eta')\nu(\eta')} \leq \max\{r(\eta), r(\eta')\}.$$

It follows that

$$\mathcal{E}_{\mu_\tau}(f, f) \leq e^{D_\tau} \mathcal{E}_\nu(f, f). \quad (2.4)$$

Because f depends only on the interior, disintegration and total variance give

$$\mathcal{E}_\pi(f, f) = \mathbb{E}_\pi \mathcal{E}_{\mu_\tau}(f, f), \quad \text{Var}_\pi(f) \geq \mathbb{E}_\pi \text{Var}_{\mu_\tau}(f).$$

Combining these identities with (2.3)–(2.4) bounds this function's Rayleigh quotient by

$$\text{gap}(\nu) \frac{\mathbb{E}_\pi e^{D_\tau}}{\mathbb{E}_\pi e^{-D_\tau}} = \text{gap}(\nu) e^{4A_B} \frac{\mathbb{E}_\pi e^{bk(\eta_S)}}{\mathbb{E}_\pi e^{-bk(\eta_S)}},$$

which proves (2.2). □

Lemma 2.3 (Equilibrium shell defects). *Suppose $\zeta_y h_y \geq H$ for every $y \in S$, and put*

$$q_H = \frac{1}{1 + e^{2(H-2d\beta)}}.$$

Then

$$\mathbb{E}_\pi e^{bk(\eta_S)} \leq e^{|S|q_H(e^b - 1)}, \quad \mathbb{E}_\pi e^{-bk(\eta_S)} \geq e^{-b|S|q_H}. \quad (2.5)$$

In particular,

$$\text{gap}(\pi) \leq \text{gap}(\nu) \exp\left\{4A_B + |S|q_H(e^b - 1 + b)\right\}. \quad (2.6)$$

Proof. For every conditioning of the other spins, the local field in the ζ_y direction is at least $H - 2d\beta$. Hence the conditional probability of a defect at y is at most q_H . This remains true after conditioning on any subset of the other spins, by averaging the full conditional probability. Exposing shell spins in a fixed order and using independent uniforms therefore couples their defect indicators below independent $\text{Ber}(q_H)$ variables. Thus

$$\mathbb{E}_\pi e^{bk(\eta_S)} \leq (1 + q_H(e^b - 1))^{|S|} \leq e^{|S|q_H(e^b - 1)}.$$

Also $\mathbb{E}_\pi k(\eta_S) \leq |S|q_H$, so Jensen's inequality yields $\mathbb{E}_\pi e^{-bk(\eta_S)} \geq e^{-b|S|q_H}$. Equation (2.6) follows from Lemma 2.2. $\square$

The numerator in (2.2) measures the possible increase of the Dirichlet form, while the denominator controls the variance remaining in the pure-system mode. Their ratio therefore accounts for defective shell configurations without requiring a defect-free trajectory. A small, fixed equilibrium defect density retains a positive fraction of the surface barrier.

We now choose the island so that the two perturbation terms in (2.6) consume at most half the pure-system barrier. The required shell threshold will be independent of the island size; only the interior fields shrink with m . Lemma A.1 then converts the spectral estimate to the claimed mixing lower bound. A surface-exponential pure-box mixing lower bound would equally suffice as input, since $\log(1/\nu_{\min}) = O(m^d)$ makes the conversion back to a gap cost only a polynomial factor.

Proof of Proposition 2.1. Let $b_0 = 8d\beta$, choose $\epsilon_0 = c_0/16$, and fix $H_0 > 0$ so large that

$$2dq_{H_0}(e^{b_0} - 1 + b_0) \leq \frac{c_0}{4}. \quad (2.7)$$

These constants do not depend on m or n . For a translate B of B_m , with $m \in \mathcal{M}$, let $S = \partial_{\text{ext}} B$ and translate ζ_m to S . Consider the event

$$\mathcal{G}_B = \left\{ |h_x| \leq \frac{\epsilon_0}{m} \ (x \in B), \quad \zeta_{m,y} h_y \geq H_0 \ (y \in S) \right\}. \quad (2.8)$$

If $B \cup S \subset \Lambda_n$ and $\mathcal{G}_B$ occurs, then $A_B \leq \epsilon_0 m^{d-1}$ and $|S| = 2dm^{d-1}$. For every sufficiently large admissible m , (1.3), (2.6), and (2.7) give

$$\text{gap}_{\Lambda_n}(h) \leq \exp\left\{-c_0 m^{d-1} + 4\epsilon_0 m^{d-1} + 2dm^{d-1}q_{H_0}(e^{b_0} - 1 + b_0)\right\} \leq e^{-(c_0/2)m^{d-1}}. \quad (2.9)$$

The fields outside $B \cup S$ are unrestricted. In particular, one occurrence of $\mathcal{G}_B$ yields a slow mode of the full chain, regardless of the disorder at the other candidate locations.

For large m , the Gaussian density on a fixed neighborhood of zero gives $\mathbb{P}_h(|h_0| \leq \epsilon_0/m) \geq c_\sigma/m$. The one-sided probability $p_{H_0,\sigma} = \mathbb{P}_h(h_0 \geq H_0)$ is a positive constant; by symmetry it is also the probability of $h_0 \leq -H_0$. Independence therefore implies

$$-\log \mathbb{P}_h(\mathcal{G}_B) \leq m^d(\log m + C) + Cm^{d-1} \leq C_2 m^d \log m. \quad (2.10)$$

There are order $(n/m)^d$ disjoint candidate locations. To make the expected number of successful islands diverge while retaining the largest useful barrier, choose a sufficiently small fixed $\gamma > 0$ and set

$$r_n = \left(\gamma \frac{\log n}{\log \log n} \right)^{1/d}, \quad m = m(n) = \max\{m_k \in \mathcal{M} : m_k \leq r_n\}. \quad (2.11)$$

The density condition on $\mathcal{M}$ gives $m/r_n \rightarrow 1$, so

$$m^d \log m = \left(\frac{\gamma}{d} + o(1) \right) \log n.$$

Decreasing γ if necessary, (2.10) yields $\mathbb{P}_h(\mathcal{G}_B) \geq n^{-d/2}$ for all large n . There are at least $J_n \geq c_d(n/m)^d$ translates whose sets $B \cup S$ are contained in Λ_n and pairwise disjoint. Their island events are independent. Thus

$$\mathbb{P}_h(\text{no such island}) \leq \exp\{-J_n n^{-d/2}\} \leq \exp\{-n^{d/2-o(1)}\}.$$

On the complementary event, (2.9) and Lemma A.1 apply. Finally,

$$m^{d-1} = (\gamma^{(d-1)/d} + o(1)) \left(\frac{\log n}{\log \log n} \right)^{(d-1)/d}.$$

Absorb the fixed factor $\log 2$ from (A.1) by decreasing the exponent constant. $\square$

The island argument extends beyond Gaussian fields. If an i.i.d. field law satisfies $\mathbb{P}(|h_0| \leq u) \geq cu^\alpha$ for some $c, \alpha > 0$ and all sufficiently small $u > 0$, and both tails $\mathbb{P}(h_0 \geq H_0)$ and $\mathbb{P}(h_0 \leq -H_0)$ are positive for a threshold satisfying (2.7), then (2.10) still holds with a different constant. The same proof gives Proposition 2.1 for that law. Only these small-ball and shell-tail probabilities enter the lower bound.

3 Worst-start mixing at strong disorder

The upper bound depends on a balance between updates that remove a discrepancy and updates that spread it. If two configurations differ only at y , resampling a block containing y can erase the difference. A block with y on its boundary can instead create new differences inside. We construct a covering in which the number I_y of the former blocks is of order R^d , the number J_y of the latter is at most order R^{d-1} , and one boundary perturbation creates at most $H_R = o(R)$ expected disagreements. Thus $I_y - H_R J_y$ stays positive. The remaining task is to realize this contraction using single-site updates in a time comparable to the slowest local block chain.

Proposition 3.1 (Worst-start upper bound). *Under Assumption 1.2, there is $C > 0$ such that, with $\mathbb{P}_h$ -probability tending to one,*

$$\max\{t_{\text{rel}}(n, h), t_{\text{mix}}(1/4; n, h)\} \leq \exp\left\{C(\log n)^{(d-1)/d}(\log \log n)^{d-1}\right\}. \quad (3.1)$$

3.1 Constructing the block covering

A weak-field site may transmit a disagreement; at a strong-field site, a suitable coupling fails to fix the preferred spin with probability at most ρ . The following auxiliary percolation records both possibilities. Its quenched cluster tails will identify suitable locations for block boundaries.

Fix K and ρ as in (1.4). Independently of h , let $(u_x)_{x \in \mathbb{Z}^d}$ be independent uniforms on $[0, 1]$, and set

$$\omega_x = \mathbf{1}_{\{|h_x| \leq K\}} \vee \mathbf{1}_{\{u_x > 1-\rho\}}. \quad (3.2)$$

Let C_z be the nearest-neighbor open cluster of z , with $C_z = \emptyset$ when $\omega_z = 0$. Write $\mathbb{P}_u$ for the probability over the uniforms at fixed field.

Take a sufficiently large integer R divisible by 8, and put $s = R/8$ and $r_0 = \lceil (\log R)^2 \rceil$, with $r_0 \leq s$. Write $Q_r(z) = \{x \in \mathbb{Z}^d : \|x - z\|_\infty \leq r\}$ for lattice cubes. Call a coarse site $v \in R\mathbb{Z}^d$ good if

$$\mathbb{P}_u(|C_z| \geq r \mid h) \leq e^{-r} \quad \text{for every } z \in Q_R(v) \text{ and every integer } r_0 \leq r \leq s. \quad (3.3)$$

Otherwise call it bad. These labels are quenched: they depend only on h , after averaging over u . Denote the good and bad coarse sites by $\mathbf{G}_R$ and $\mathbf{B}_R$. Two coarse sites are *star-adjacent* if their ℓ^∞ -distance is R . Bad clusters are the connected components of $\mathbf{B}_R$ for this adjacency.

Lemma 3.2 (Size of bad clusters). *There are $R_0, \kappa > 0$ such that, whenever $R_0 \leq R \leq n$, $8 \mid R$, and $L \geq \kappa \log n$, the following event has probability at least $1 - e^{-L}$: every bad cluster containing a coarse site v with $\|v\|_\infty \leq n + R$ has at most L sites.*

Proof. Under the joint law of h, u , the marks in (3.2) are independent Bernoulli variables of parameter

$$p = p_K + (1 - p_K)\rho < \frac{1}{20d}.$$

A cluster exploration that discovers r open sites exposes at most $2dr$ independent marks before the r -th discovery. Padding a stopped exploration by unused independent marks gives

$$\mathbb{P}_{h,u}(|C_z| \geq r) \leq \mathbb{P}(\text{Bin}(2dr, p) \geq r) \leq (2dep)^r \leq (e/10)^r.$$

For the middle inequality, exponential Markov bounds the binomial upper tail by $(emp/r)^r$, with $m = 2dr$. Applying Markov's inequality to the conditional probability and then summing over z, r shows that

$$q_R := \sup_v \mathbb{P}_h(v \in \mathbf{B}_R) \leq CR^d \sum_{r=r_0}^s (e^2/10)^r \leq CR^d e^{-cr_0}, \quad (3.4)$$

where $c = \log 10 - 2 > 0$. In particular, $q_R \rightarrow 0$.

The event $\{|C_z| \geq r\}$ is determined by the marks within graph distance $r - 1$ of z : if it occurs, an open connected set of exactly r vertices containing z can be found there. Thus the label of v depends only on the field in $Q_{2R}(v)$. Partitioning $R\mathbb{Z}^d$ by its coordinate residues modulo $5R$ gives 5^d classes, within each of which the labels are independent. Every set of ℓ coarse sites contains at least $\ell/5^d$ sites in one such class. The probability that all ℓ sites are bad is therefore at most $q_R^{\ell/5^d}$.

Let $\Delta_c = 3^d - 1$. There are at most $\Delta_c^{2\ell}$ star-connected sets of size ℓ containing a given coarse site: choose a spanning tree of each set and encode it by a depth-first walk of length $2(\ell - 1)$. For all sufficiently large R , $\Delta_c^2 q_R^{1/5^d} \leq e^{-3}$. Since a cluster of size at least ℓ contains a connected subset of exactly that size through its root,

$$\mathbb{P}_h(\text{the bad cluster at } v \text{ has size at least } \ell) \leq e^{-3\ell}.$$

This also excludes infinite bad clusters almost surely. There are at most $C_d n^d$ relevant coarse sites when $R \leq n$. A union bound gives failure probability at most $C_d n^d e^{-3L} \leq e^{-L}$, on choosing κ large enough. $\square$

The bad clusters determine the regions that a block must absorb. The block family also needs overlap: translating each candidate block gives a site many opportunities to be updated in its interior, while placing it on the boundary of relatively few translates. We implement these two requirements using disjoint lattice cells.

Partition $\mathbb{Z}^d$ into the cells

$$T_v = v + \{-R/2, -R/2 + 1, \dots, R/2 - 1\}^d, \quad v \in R\mathbb{Z}^d.$$

For a bad cluster $\mathcal{C}$, set $U_{\mathcal{C}} = \bigcup_{v \in \mathcal{C}} T_v$, and let $\mathcal{A}_s = \{-s, -s+1, \dots, s\}^d$. We use two types of blocks in $V = \Lambda_n$. For each lattice point w such that $\text{dist}_{\infty}(w, \mathbf{G}_R) \leq 3R/4$, include $Q_s(w) \cap V$ when it is nonempty. For each bad cluster $\mathcal{C}$ and each $a \in \mathcal{A}_s$, include $(U_{\mathcal{C}} + a) \cap V$ when it is nonempty. The resulting indexed family is $\mathcal{B} = (B_i)_i$. Repeated sets are retained as distinct indices.

Only finitely many blocks meet V , almost surely. A type-1 center lies within distance s of V . A type-2 block meeting V comes from a cluster containing a coarse site of norm at most $n + R/2 + s \leq n + R$. The extension of the disorder to $\mathbb{Z}^d$ merely supplies auxiliary randomness for this construction; the Gibbs measure and dynamics in V still use only $h|_V$.

Lemma 3.3 (Coverage and boundary counts). *On the event in Lemma 3.2, put*

$$M = \max_i |B_i|, \quad I_y = |\{i : y \in B_i\}|, \quad J_y = |\{i : y \in \partial_V B_i\}|.$$

For all sufficiently large R ,

$$M \leq LR^d, \quad cR^d \leq I_y \leq CR^d, \quad J_y \leq CR^{d-1} \quad (y \in V). \quad (3.5)$$

Every $y \in \partial_V B_i$ lies in $Q_R(v)$ for some good coarse site v .

Proof. A type-1 block has at most $(2s+1)^d \leq R^d$ vertices. A type-2 block has at most $|\mathcal{C}|R^d \leq LR^d$ vertices, by the choice of the event.

Fix $y \in V$. If $\text{dist}_{\infty}(y, \mathbf{G}_R) \leq 5R/8$, every center $w \in Q_s(y)$ qualifies for a type-1 block, since $\text{dist}_{\infty}(w, \mathbf{G}_R) \leq 3R/4$. All $(2s+1)^d$ such blocks contain y , including when y is on the boundary of V . Otherwise, every coarse cell meeting $Q_s(y)$ is bad. Indeed, a point in such a cell is within s of y and within $R/2$ of its center, so a good center would contradict the assumed distance. Since $2s < R$, the centers of these cells are pairwise equal or star-adjacent. They belong to one bad cluster $\mathcal{C}$, and $Q_s(y) \subset U_{\mathcal{C}}$. Consequently y belongs to all $(2s+1)^d$ translates from this family. This proves the lower bound on I_y .

Cells from distinct bad clusters have ℓ^{∞} -distance at least $R+1$: their centers cannot be star-adjacent. After shifts in $\mathcal{A}_s$, the distance is at least $R+1-2s$. For large R , even their one-step closures are disjoint. Thus at most one type-2 family can have y in a block or its boundary. At most $(2s+1)^d$ blocks of each type contain y , proving the upper bound on I_y .

A type-1 block with y on its external vertex boundary has a center at distance $s+1$ in one coordinate and at most s in every other coordinate. There are at most $2d(2s+1)^{d-1}$ such centers. For type 2, if $y \in \partial_V((U_{\mathcal{C}} + a) \cap V)$, then $y - a \in \partial_{\text{ext}} U_{\mathcal{C}}$. In at least one coordinate j , this point is on the first or last lattice layer of a coarse cell. Hence $y_j - a_j$ is congruent to either $-R/2$ or $R/2 - 1$ modulo R . For each j , at most two choices of $a_j \in [-s, s]$ satisfy these conditions, and the remaining coordinates have at most $(2s+1)^{d-1}$ choices. Since only one cluster family can contribute, this gives at most $2d(2s+1)^{d-1}$ type-2 boundaries through y . The bound on J_y follows.

Finally, a type-1 boundary point is within $3R/4 + s + 1 \leq R$ of a good center. For type 2, take adjacent $y - a \notin U_{\mathcal{C}}$ and $x - a \in U_{\mathcal{C}}$. The cell containing $y - a$ is adjacent to a cell of $\mathcal{C}$. Its center v must be good, since a bad center would belong to $\mathcal{C}$. Thus $\|y - v\|_{\infty} \leq R/2 + s \leq R$. Intersecting with V introduces no additional case: a relative boundary point y is itself in V , so it was already outside the untruncated block. $\square$

3.2 Coupling a boundary perturbation

Every block boundary lies near a good coarse site, even when the block encloses a cluster of weak fields. To use this fact, the coupling must work on the entire, possibly irregular block. We expose a disagreement cluster until strong-field sites seal it off; no inscribed-box argument is required.

Let d_H be Hamming distance, and let W_H be the corresponding Wasserstein distance, namely the infimum of $\mathbb{E}d_H(X, Y)$ over couplings of the specified laws.

Lemma 3.4 (Disagreement-percolation estimate). *For every block constructed above, every $y \in \partial_V B_i$, and any two boundary conditions ϕ, ϕ^y differing only at y ,*

$$W_H(\mu_{B_i, h}^\phi, \mu_{B_i, h}^{\phi^y}) \leq C(\log R)^2 + C|B_i|e^{-R/8}. \quad (3.6)$$

The constants are uniform over the block and boundary conditions.

Proof. We first construct a coupling on an arbitrary finite domain $D \subset V$ whose boundary conditions differ at y . If $|h_x| > K$, then, under any conditioning of the other spins,

$$\mu_{D, h}^\phi(\eta_x \neq \text{sgn}(h_x) \mid \text{the conditioned spins}) \leq \rho. \quad (3.7)$$

For full conditioning this follows because the interaction part of the local field has magnitude at most $2d\beta$. For partial conditioning it follows by averaging the full conditional bound. It holds for both boundary conditions.

Force y open, and start exploring from its neighbors in D . Choose $+1$ as the preferred spin when $h_x = 0$. At each exposed vertex, use the same independent uniform u_x to sample the two conditional one-site laws, assigning the interval $[0, p_x]$ to the preferred spin in each marginal, where p_x is that marginal's conditional probability of the preferred spin. By (3.7), at a strong-field vertex both spins equal $\text{sgn}(h_x)$ whenever $u_x \leq 1 - \rho$. Expose the unexamined neighbors of every open vertex, with openness as in (3.2); do not continue through a closed vertex. At a weak-field vertex the mark is always open. A fixed tie-breaking order makes this exploration unambiguous.

At termination, every exposed closed vertex has the same spin in the two samples. These vertices, together with the unchanged original boundary, separate the unexplored region from the boundary perturbation. The conditional Gibbs measures on the unexplored region are therefore identical, and can be sampled identically. Sampling successive conditional marginals and then this common conditional law gives the required Gibbs marginals. All disagreements are confined to the open cluster reached from the forced-open root in $D \cup \{y\}$. If C_y^* is the cluster on the full lattice with only y forced open, this proves

$$d_H(\eta, \eta^y) \leq \min\{|D|, |C_y^*|\}. \quad (3.8)$$

Allowing paths outside D only enlarges the dominating cluster. In particular, this construction imposes no condition on the shape of D .

The perturbation at the root is imposed by the two boundary conditions, so the root must be forced open in the dominating percolation. Given h , its ordinary opening probability is $p_y \in \{\rho, 1\}$. Independence of the marks at fixed field gives, for every integer $r \geq 1$,

$$\mathbb{P}_u(|C_y^*| \geq r \mid h) = \mathbb{P}_u(|C_y| \geq r \mid h, \omega_y = 1) = \frac{\mathbb{P}_u(|C_y| \geq r \mid h)}{p_y}. \quad (3.9)$$

For $D = B_i$, Lemma 3.3 places y in the neighborhood of a good coarse site. Thus (3.3) and (3.9) bound this probability by $\rho^{-1}e^{-r}$ for $r_0 \leq r \leq s$. Summing the integer tails in (3.8) gives

$$\mathbb{E}d_H(\eta, \eta^y) \leq r_0 + \rho^{-1} \sum_{r=r_0}^s e^{-r} + \rho^{-1}|B_i|e^{-s} \leq C(\log R)^2 + C|B_i|e^{-R/8}.$$

The factor ρ^{-1} is a fixed constant, independent of R, L, n . This proves (3.6). $\square$

The influence estimate must be small relative to the coverage ratio I_y/J_y , which is of order R . We therefore choose

$$L = \lceil \kappa \log n \rceil, \quad R = 8 \left\lceil \frac{C_1 \log L}{8} \right\rceil, \quad N = |\Lambda_n|, \quad (3.10)$$

with κ as in Lemma 3.2 and $C_1 > 16$. These choices satisfy that lemma's conditions for all large n . On its event, let

$$H_R = C(\log R)^2 + CM e^{-R/8}.$$

Then (3.5) and (3.10) imply

$$H_R = o(R), \quad \frac{J_y}{I_y} \leq \frac{C}{R}. \quad (3.11)$$

Indeed, $Me^{-R/8} \leq R^d L^{1-C_1/8} = o(R)$, and $(\log R)^2 = o(R)$.

The same blocks have a local relaxation estimate that is uniform in the fields. Applying Lemma A.2 with $|B_i| \leq LR^d$ gives

$$\inf_{i,\phi} \text{gap}(\mu_{B_i,h}^\phi) \geq C^{-1} \exp\{-CR^{d-1}L^{(d-1)/d}\}. \quad (3.12)$$

This is where lattice geometry enters the time scale: the cutwidth bound in Lemma A.2 charges surface order in the block volume, while its field cancellation permits arbitrary exterior pinnings. With $L \asymp \log n$ and $R \asymp \log \log n$, the exponent in (3.12) is exactly the one needed in Proposition 3.1.

3.3 From block contraction to single-site mixing

Fix a field realization for which the block estimates hold. Sequentially simulating a sweep through all blocks would charge for the size of the entire system. Instead, partition the blocks into classes with disjoint closures, and update all blocks of a chosen class simultaneously. Only polylogarithmically many classes are needed because both block size and overlap are polylogarithmic.

Let $\bar{B}_i = B_i \cup \partial_V B_i$, and join two block indices when their closures intersect. From (3.5), at most $I_y + J_y \leq CR^d$ closures contain any site, and $|\bar{B}_i| \leq (2d+1)M$. The conflict graph therefore has maximum degree at most CMR^d . A greedy coloring uses

$$Q \leq CMR^d \quad (3.13)$$

colors, with disjoint closures within each color. Fix one such coloring as a function of the field.

An ideal step chooses a color uniformly and resamples every block of that color from its conditional Gibbs measure. These resamplings factorize: no edge joins distinct blocks of the same color. Let K denote the ideal-step kernel.

Lemma 3.5 (Ideal-step contraction). *For all sufficiently large n ,*

$$W_H(K\eta, K\eta') \leq (1 - c/M)d_H(\eta, \eta') \quad (3.14)$$

for every pair of configurations.

Proof. First suppose the configurations differ only at y . Use the same color in both chains. If a selected block contains y , its two boundary conditions agree, and a common Gibbs sample removes the discrepancy. If y lies on a selected block's boundary, its original discrepancy remains and Lemma 3.4 bounds the expected added discrepancies by H_R . All unaffected blocks can be coupled identically. Disjoint closures ensure that at most one block is affected for each chosen color. Hence

$$W_H(K\eta, K\eta') \leq 1 - \frac{I_y}{Q} + \frac{H_R J_y}{Q}. \quad (3.15)$$

By (3.11), $I_y - H_R J_y \geq I_y/2$ for large n , and (3.5)–(3.13) give $I_y/Q \geq c/M$. This proves the one-discrepancy bound. For a general pair, apply the triangle inequality for W_H along a shortest Hamming path between the two initial configurations. $\square$

The ideal kernel contracts, but its exact resampling step is not an update of Glauber dynamics. We next replace it by a local chain run long enough to approximate each conditional law. The minimum stationary mass enters only at this stage, on a set of at most M sites. The additional field event

$$\max_{x \in V} |h_x| \leq \sigma \sqrt{8 \log N} \quad (3.16)$$

has probability at least $1 - N^{-2}$ for $N \geq 2$, by the Gaussian tail bound and a union bound. On this event, every block and every boundary condition satisfy

$$\log \frac{1}{\mu_{B_i, h, \min}^\phi} \leq CM(1 + \sqrt{\log N}). \quad (3.17)$$

To see this, the logarithm of the number of configurations is at most $M \log 2$; the oscillation of the interaction energy is $O(\beta M)$, and that of the field energy is at most $2M \max_{x \in V} |h_x|$.

For a pair initially differing at one site, a selected color has at most one affected block. This allows us to compare the two local outputs only on that block, rather than summing approximation errors over every block in the system. Choose $\delta_M = c_*/M^2$, with $c_* > 0$ fixed and small. Lemma A.1, together with (3.12) and (3.17), shows that

$$T = C \exp\{CR^{d-1}L^{(d-1)/d}\} \left[M(1 + \sqrt{\log N}) + 2 \log M + 1 \right] \quad (3.18)$$

can be chosen so that every block chain, with its exterior frozen, is within δ_M of equilibrium at time T , uniformly in its initial configuration and boundary condition.

Replace an ideal step by running these rate-one local dynamics for time T in the blocks of the chosen color. They run in parallel, so one step costs physical time T , not T times the number of selected blocks. Let $\tilde{K}$ denote the resulting kernel.

Lemma 3.6 (Simulated-step contraction). *For sufficiently small fixed c_* and all large n ,*

$$W_H(\tilde{K}\eta, \tilde{K}\eta') \leq (1 - c'/M)d_H(\eta, \eta'). \quad (3.19)$$

Proof. Consider again a pair differing only at y . For an affected block, couple each of its two local outputs to its equilibrium sample with failure probability at most δ_M , and couple the equilibrium samples as in the ideal step. Gluing these couplings adds at most $2M\delta_M$ to the expected Hamming cost. Unaffected blocks have identical initial states and boundary conditions, so their actual local evolutions can be coupled identically, without any approximation cost. Averaging over the chosen color gives

$$W_H(\tilde{K}\eta, \tilde{K}\eta') \leq 1 - \frac{I_y}{Q} + \frac{H_R J_y}{Q} + \frac{2M\delta_M(I_y + J_y)}{Q}.$$

The first three terms are at most $1 - I_y/(2Q)$. Since $I_y + J_y \leq CI_y$, the last term is at most $Cc_*I_y/(MQ)$, and hence at most $I_y/(4Q)$ when c_* is sufficiently small. Use $I_y/Q \geq c/M$ and then the Hamming-path argument from Lemma 3.5. $\square$

Write ν_k^- and ν_k^+ for the laws after k simulated steps from the all-minus and all-plus states. Iterating (3.19) gives

$$W_H(\nu_k^-, \nu_k^+) \leq N(1 - c'/M)^k < \frac{1}{8} \quad \text{for } k = \lceil CM \log(16N) \rceil, \quad (3.20)$$

with a sufficiently large fixed C . The extension of the kernel contraction to probability laws follows by integrating over a coupling of the input laws.

It remains to transfer the estimate from the simulated kernel to the original chain. The simulation is a censoring: during each interval of length T , retain only clock rings in the union of the chosen color's blocks. At fixed field, the color choices and Poisson clocks are independent of the spin updates. Conditional on this randomness, the retained sequence is a prescribed subsequence of the full, almost surely finite update sequence. The censoring inequality [5, Theorem 1.1], and its order-reversed version, can therefore be applied before averaging over colors and clocks. At time $t = kT$, they give

$$\nu_k^- \preceq P_t^h(-, \cdot) \preceq P_t^h(+, \cdot) \preceq \nu_k^+. \quad (3.21)$$

Here $\preceq$ denotes stochastic order; the middle inequality is attractiveness of ferromagnetic heat-bath dynamics.

For two ordered spin laws $\alpha^- \preceq \alpha^+$,

$$W_H(\alpha^-, \alpha^+) = \frac{1}{2} \sum_{x \in V} (\mathbb{E}_{\alpha^+} \eta_x - \mathbb{E}_{\alpha^-} \eta_x). \quad (3.22)$$

Every coupling has at least this expected coordinatewise cost, and a monotone coupling attains it. Thus (3.20)–(3.21) imply that the full top and bottom chains have Wasserstein distance less than $1/8$. In their monotone grand coupling, the expected Hamming distance equals this Wasserstein distance, by (3.22). A chain from any initial state and a stationary chain can be placed between the same extremes. The coupling inequality then gives, uniformly in η ,

$$\|P_t^h(\eta, \cdot) - \mu_{V,h}\|_{\text{TV}} \leq \mathbb{P}(X_t^+ \neq X_t^-) \leq \mathbb{E} d_H(X_t^+, X_t^-) < \frac{1}{8}.$$

Proof of Proposition 3.1. The preceding argument bounds $t_{\text{mix}}(1/4; n, h)$ by kT outside an event of probability at most $e^{-L} + N^{-2}$. Equations (3.5), (3.18), and (3.20) give

$$\log(kT) \leq CR^{d-1}L^{(d-1)/d} + O(\log \log N).$$

Insert (3.10) to obtain the mixing-time bound. The relaxation-time bound follows from the first inequality in (A.1), after adjusting the constant. Although the block event was constructed using the field on the full lattice, both quantities in (3.1) depend only on $h|_V$. The same probability bound therefore holds for its marginal law. $\square$

Proof of Theorem 1.3. Intersect the events of Propositions 2.1 and 3.1. This proves (1.5) for both quantities simultaneously. On this event, for either choice of Y_n ,

$$a \log \log n - a \log \log \log n + O(1) \leq \log \log Y_n \leq a \log \log n + (d-1) \log \log \log n + O(1).$$

Dividing by $\log \log n$ proves (1.6). $\square$

A Spectral estimates

A.1 Relaxation and total-variation mixing

For a finite irreducible reversible continuous-time chain with at least two states and stationary law π , write $\pi_{\min} = \min_x \pi(x)$. The following estimates use the same time normalization as the chain under consideration.

Lemma A.1. *The relaxation and mixing times satisfy*

$$t_{\text{mix}}(1/4) \geq (\log 2) \text{gap}^{-1}, \quad t_{\text{mix}}(\varepsilon) \leq \text{gap}^{-1} \log \frac{1}{\varepsilon \pi_{\min}} \quad (0 < \varepsilon < 1/2). \quad (\text{A.1})$$

Proof. Normalize a mean-zero gap eigenfunction by $\|f\|_{\infty} = 1$, and choose x with $|f(x)| = 1$. Since $|P_t f(x)| = e^{-t \text{gap}}$, total-variation duality gives $\|P_t(x, \cdot) - \pi\|_{\text{TV}} \geq \frac{1}{2} e^{-t \text{gap}}$. This proves the lower bound. For the upper bound, reversibility and $L^2(\pi)$ contraction imply

$$\left\| \frac{P_t(x, \cdot)}{\pi} - 1 \right\|_{L^2(\pi)} \leq e^{-t \text{gap}} \sqrt{\pi(x)^{-1} - 1}.$$

Cauchy–Schwarz bounds total variation by half this quantity. The displayed upper bound is a convenient weakening of the resulting estimate. $\square$

A.2 A local bound uniform in fields and boundary conditions

The estimate behind (3.12) is the cutwidth bound underlying [3, Lemma 5.12]. It holds uniformly in the external fields because the canonical-path comparison pairs configurations so that all one-site energy terms cancel. Only interaction edges crossing the ordering cut remain. We include the argument for arbitrary finite sets, including disconnected ones.

Lemma A.2. *For every $d \geq 2$ and $\beta > 0$, there is $C = C(d, \beta)$ such that the rate-one heat-bath dynamics on any nonempty finite $A \subset \mathbb{Z}^d$, with arbitrary finite external fields and arbitrary fixed boundary conditions, satisfies*

$$\text{gap}_A \geq C^{-1} \exp\{-C|A|^{(d-1)/d}\}. \quad (\text{A.2})$$

Proof. Write $q = |A|$ and $a = (d-1)/d$. We first produce an ordering of A with at most $C_d q^a$ edges crossing any initial-segment cut. For sufficiently large q , let $\ell = \lfloor (q/2)^{1/d} \rfloor \geq 2$. Choose independent uniform residues modulo ℓ in the d coordinates, and remove the set $S \subset A$ of vertices lying on at least one of the corresponding coordinate hyperplanes. Each vertex is removed with probability at most d/ℓ , so some choice has

$$|S| \leq dq/\ell \leq C_d q^a.$$

Every remaining component is contained in a cell of side $\ell - 1$, and hence has at most $q/2$ vertices. Order these components recursively, one after another, and place S last. Before S is reached, crossing edges consist of edges incident to S , at most $2d|S|$, together with crossing edges in the currently ordered component. While S itself is ordered, all crossing edges are incident to S . Along a recursion branch, component sizes decrease by a factor of at least two. The geometric sum $\sum_{j \geq 0} (q/2^j)^a$ is $O_d(q^a)$. Treating bounded q directly proves the claimed cutwidth bound; denote it by $w \leq C_d q^a$.

Absorb fixed boundary spins into the external fields, and call the Gibbs measure μ . For every ordered pair of configurations (α, γ) , use the path that changes their differing coordinates in the chosen vertex order. Its length is at most q . Consider a directed path edge $z \rightarrow z'$ changing coordinate i . With subscripts referring to the chosen ordering,

$$z = (\gamma_{<i}, \alpha_{\geq i}), \quad z' = (\gamma_{\leq i}, \alpha_{>i}), \quad \xi = (\alpha_{<i}, \gamma_{\geq i}), \quad \xi' = (\alpha_{\leq i}, \gamma_{>i}).$$

In the product comparison between (α, γ) and (z, ξ) , all one-site field terms cancel, as do interaction terms not crossing the cut before i . Each crossing edge changes the logarithmic weight by at most 4β . The analogous statement holds for (z', ξ') at the cut after i . Consequently,

$$\mu(\alpha)\mu(\gamma) \leq e^{4\beta w} \mu(z)\mu(\xi), \quad \mu(\alpha)\mu(\gamma) \leq e^{4\beta w} \mu(z')\mu(\xi').$$

The heat-bath conductance is $c(z, z') = \mu(z)\mu(z')/(\mu(z) + \mu(z'))$, so

$$\frac{\mu(\alpha)\mu(\gamma)}{c(z, z')} \leq e^{4\beta w}(\mu(\xi) + \mu(\xi')).$$

For a fixed directed edge, each of the maps from a path pair to ξ and to ξ' is injective: the edge determines the updated coordinate and its two values, while the complementary configuration determines the remaining coordinates of the pair. Summing over path pairs that use this edge therefore gives at most $2e^{4\beta w}$ after division by its conductance.

Finally, write variance as $\frac{1}{2} \sum_{\alpha, \gamma} \mu(\alpha)\mu(\gamma)(f(\alpha) - f(\gamma))^2$, telescope each difference along its path, and apply Cauchy–Schwarz with the path length. Summing the resulting edge contributions yields

$$\text{Var}_\mu(f) \leq 2qe^{4\beta w} \mathcal{E}_\mu(f, f).$$

Since $w \leq C_d q^a$ and $a > 0$, the factor $2q$ can be absorbed into $C \exp(Cq^a)$. This proves (A.2), uniformly in all the fields. $\square$

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